

Optics of water

Y24 (2024) — English translation

Introduction

If you have gone swimming in a river at least once, you have most likely noticed the interesting optical properties of “real” water. We are talking mainly about the processes of scattering and absorption on the water itself and the structures inside it (particles, bubbles, etc.) [see. rice. 1].



Fig. 1.

These phenomena can tell an experienced researcher a lot about the properties and characteristics of water in a particular body of water, therefore methods for solving problems of scattering and absorption are also of applied interest. This is what this task will focus on.

Part A. Water molecule and absorption coefficient (3.5 points)

The water molecule (H_2O) consists of two hydrogen atoms and one oxygen atom, forming an isosceles triangle. Like any mechanical system, a water molecule can experience vibrations around its equilibrium position. All three possible modes of vibration of a water molecule are shown in Fig. 2.

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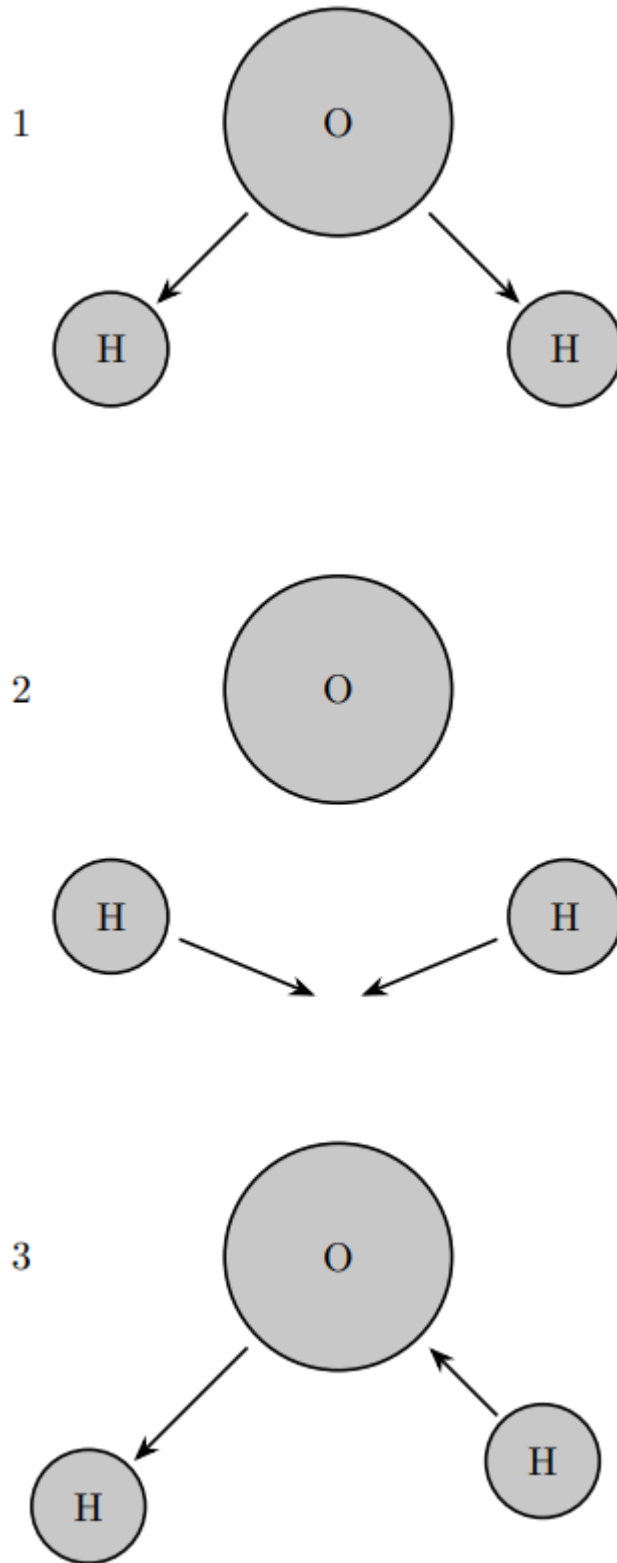


Fig. 2.

When the frequency of an electromagnetic wave propagating in water turns out to be approximately a multiple of one of the natural frequencies of vibrations of the molecule, these vibrations are excited in the water molecules with a very large amplitude, “taking away” the energy of the wave and subsequently re-emitting it in all directions. This leads to the fact that the absorption coefficient has local maxima at these frequencies [see. rice. 3]. Note: The absorption coefficient determines the distance after which the beam intensity decreases by a factor of e . Dimension $[\kappa] = \mathcal{M}^{-1}$.

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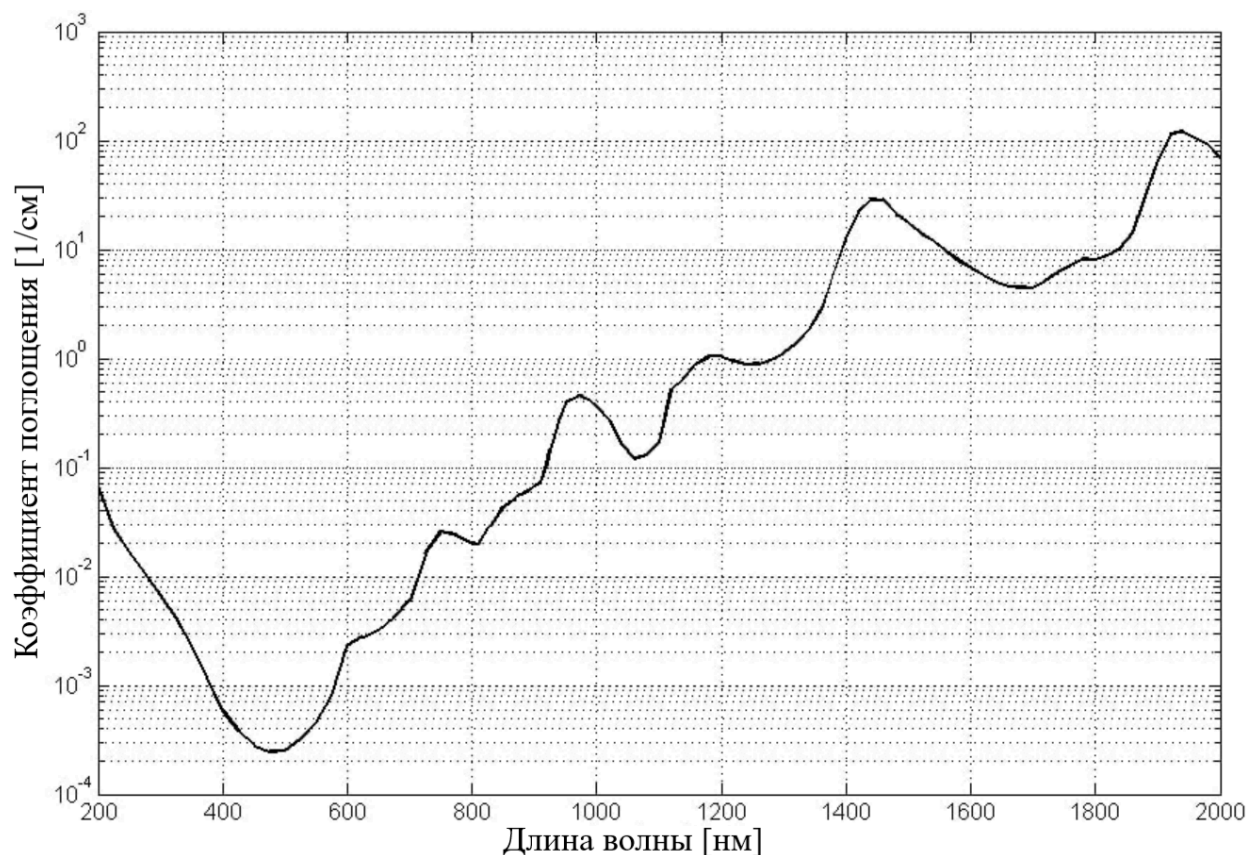


Fig. 3.

Since the mass of the oxygen atom is much greater than the mass of the hydrogen atom, it can be considered practically motionless. Moreover, hydrogen atoms interact with each other noticeably less than with the oxygen atom. According to the results of experimental studies, the frequencies of the two vibration modes practically coincide, and the frequency of the third mode turns out to be two times less than the other two with good accuracy.

A1 Indicate which mode has the lowest natural frequency. Justify your answer.

0.30

A2 Find the wavelengths λ and frequencies ν corresponding to the local maxima of the absorption coefficient.

0.90

It is clear that all frequencies corresponding to local maxima must be multiples of some fundamental frequency ν_0 . To find the fundamental frequency, it is necessary to “lay” these points on a straight line, i.e. select the frequency multiplicities so as to obtain a linear dependence. To do this, it is necessary to develop a criterion for “getting on the line.” Usually, hitting a straight line is taken as a criterion, taking into account the error cross. As an error, we take the half-width at half-height - the minimum frequency deviation at which the absorption indicator drops by 2 times.

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A3 Find the half-width $\Delta\nu$ of the local maxima of the absorption coefficient.

1.20

- A4** Based on the results obtained, determine the frequency multiplicity corresponding to the local maxima, and also find the main resonant frequency ν_0 of vibrations of the water molecule. Graphically determine the error ν_0 . **1.20**

Part B. Scattering by bubbles (6.5 points)

In “real” water, not only absorption, but also scattering of light occurs. In this problem we consider scattering by air bubbles. An example is shown in Fig. 4.



Fig. 4.

For simplicity, we consider scattering by a single bubble [see rice. 5]. Let us introduce the relative impact parameter ρ as the ratio of the impact parameter to the radius of the bubble, and denote the deflection angle as ψ . The refractive index inside the bubble is 1, outside – n .

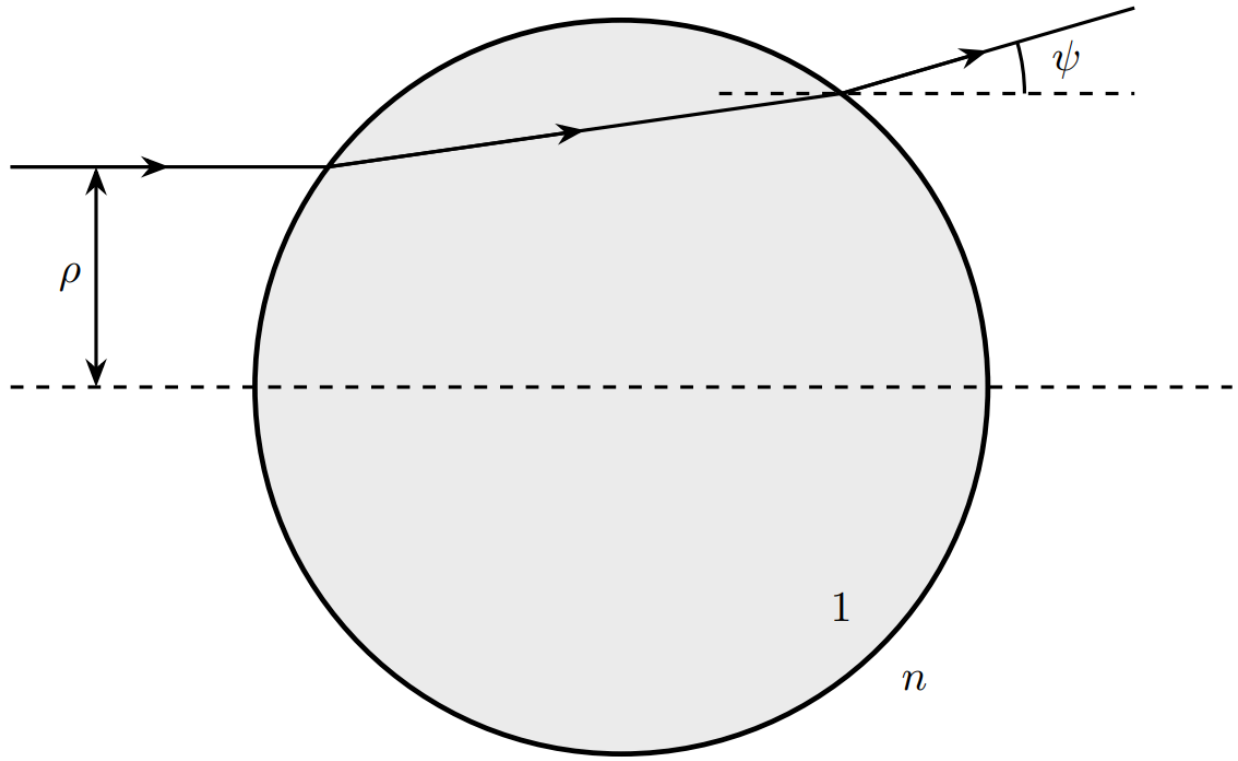


Fig. 5.

Then the deflection angle will be written as

$$\psi = 2 [\arcsin n\rho - \arcsin \rho],$$

and the intensity of the scattered radiation - as

$$\mathcal{J}(\psi) \equiv \frac{\rho \, d\rho}{\sin \psi \, d\psi}.$$

In Fig. Figure 6 shows a graph of the dependence $\mathcal{J}(\psi)$, obtained experimentally.

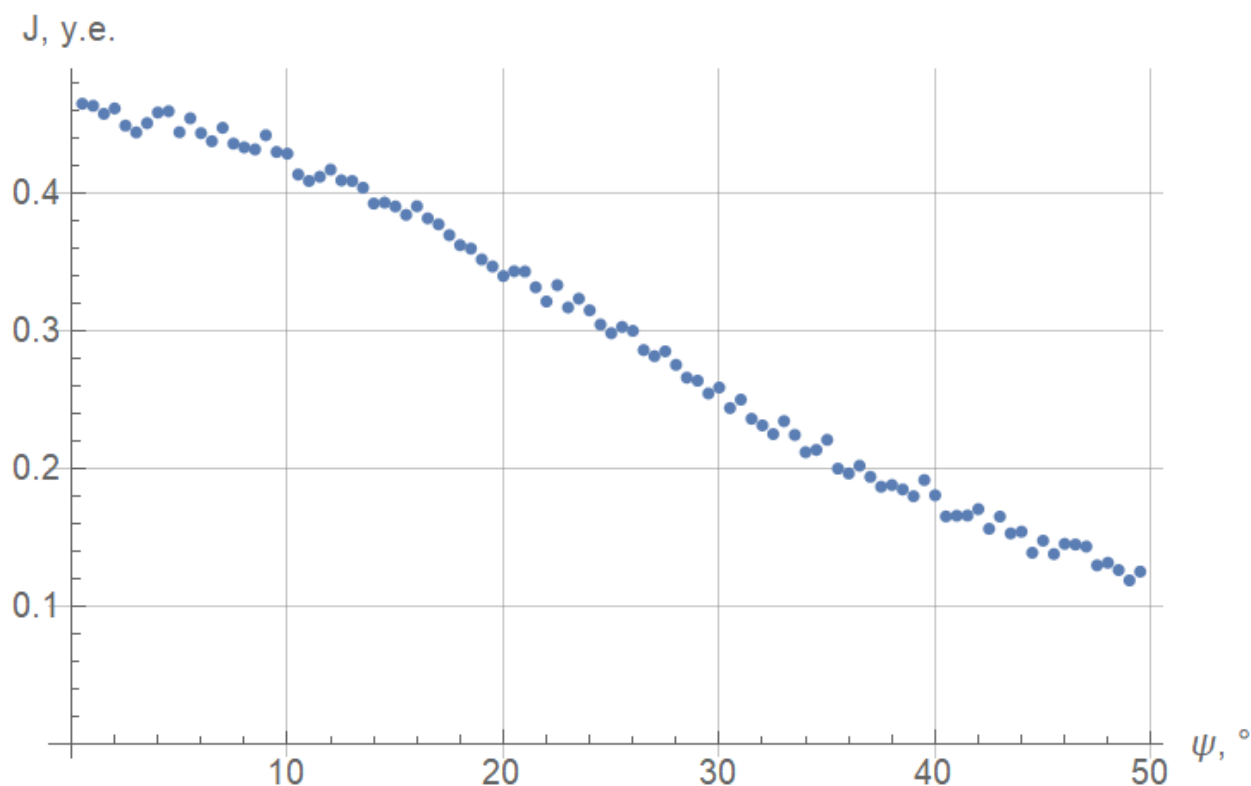


Fig. 6.

An honest solution to the inverse problem (determining the refractive index of a liquid n from the distribution $\mathcal{J}(\psi)$) is possible, but quite problematic. As a compromise, it is proposed to consider the asymptotic behavior of the curve $\mathcal{J}(\psi)$ in a “convenient” approximation.

B1 Obtain an approximate expression for $\mathcal{J}(\psi)$, considering the angles ψ small.

1.50

The table in the answer sheet shows the experimental points for $\mathcal{J}(\psi)$. $\mathcal{J}$ is given in conventional units.

$\psi, ^\circ$	$\mathcal{J}, \text{y.e.}$	$\psi, ^\circ$	$\mathcal{J}, \text{y.e.}$	$\psi, ^\circ$	$\mathcal{J}, \text{y.e.}$	$\psi, ^\circ$	$\mathcal{J}, \text{y.e.}$	$\psi, ^\circ$	$\mathcal{J}, \text{y.e.}$
0.5	0.465	5.5	0.455	1.0	0.464	6.0	0.444	1.5	0.458
2.0	0.448	7.0	0.448	2.5	0.449	7.5	0.436	3.0	0.444
3.0	0.434	8.0	0.434	3.5	0.451	8.5	0.432	4.0	0.459
4.0	0.459	9.0	0.442	4.5	0.460	9.5	0.430	5.0	0.445
5.0	0.445	10.0	0.429	10.5	0.414	15.5	0.384	11.0	0.409
11.0	0.409	16.0	0.391	11.5	0.412	16.5	0.382	12.0	0.417
12.0	0.352	12.5	0.410	17.5	0.370	13.0	0.409	18.0	0.362
13.0	0.362	13.5	0.404	18.5	0.360	14.0	0.393	19.0	0.352
14.0	0.393	14.5	0.393	19.5	0.347	15.0	0.391	20.0	0.340
20.0	0.340	20.5	0.343	25.5	0.303	21.0	0.343	26.0	0.300
21.0	0.343	26.0	0.300	21.5	0.332	26.5	0.286	22.0	0.321
22.0	0.321	27.0	0.282	22.5	0.333	27.5	0.285	23.0	0.317
23.0	0.317	28.0	0.275	23.5	0.324	28.5	0.266	24.0	0.315
24.0	0.315	29.0	0.264	24.5	0.305	29.5	0.255	25.0	0.299
25.0	0.299	30.0	0.259	30.5	0.244	35.5	0.200	31.0	0.250
31.0	0.250	36.0	0.197	31.5	0.236	36.5	0.202	32.0	0.231
32.0	0.231	37.0	0.194	32.5	0.225	37.5	0.187	33.0	0.235
33.0	0.235	38.0	0.188	33.5	0.225	38.5	0.185	34.0	0.212
34.0	0.212	39.0	0.180	34.5	0.214	39.5	0.192	35.0	0.221
35.0	0.221	40.0	0.181	40.5	0.165	45.5	0.138	41.0	0.166
41.0	0.166	46.0	0.145	41.5	0.166	46.5	0.145	42.0	0.171
42.0	0.171	47.0	0.143	42.5	0.156	47.5	0.130	43.0	0.165
43.0	0.165	48.0	0.132	43.5	0.153	48.5	0.126	44.0	0.154
44.0	0.154	49.0	0.119	44.5	0.139	49.5	0.125	45.0	0.148
45.0	0.148	—	—	—	—	—	—	—	—

B2 Linearize the resulting dependence and find the refractive index of the liquid n . Note: It is not necessary to use all points to get a good result. **5.00**

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Source: <https://pho.rs/p/3864>